Code No: 117DX **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
INFORMATION RETRIEVAL SYSTEMS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	What is Boolean Indexing?	[2]
b)	Write a brief note on PRP.	[3]
c)	Illustrate Passage-based Retrieval.	[2]
d)	Define Clustering.	[3]
e)	What is Phase Translation?	[2]
f)	Write a brief note on Parsing.	[3]
g)	What are the benefits of Signature Files?	[2]
h)	What is Index Pruning?	[3]
i)	What is a Web Search Engine?	[2]
j)	What are Proximity Searches?	[3]

PART – B

(50 Marks)

2. Consider a case insensitive query and document collection with a query Q and a document collection consisting of the following three documents:
Q: "gold silver truck"
D l : "Shipment of gold damaged in a fire"
D 2 : "Delivery of silver arrived in a silver truck"
D3: "Shipment of gold arrived in a truck"
Explain the Similarity Co-efficient for the above query and calculate the Similarity Measures. [10]

OR

3. Write a detailed note on Non-Binary Independence Model. [10]

4.a) Explain Relevance Feedback in the Vector Space Model.
b) Discuss Damashek Algorithm. [5+5]

OR

5.a) Describe hierarchical agglomerative clustering.
b) Explain about D' Amore and Mah n-gram algorithm. [5+5]

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6. Explain word net as a Semantic Networks. [10]

OR

7.a) Explain Evaluation in Cross-Language Retrieval Systems.
b) Discuss Query Translation. [5+5]

8. How can partial result set retrieval improve run-time performance of information retrieval systems? [10]

OR

9.a) Explain duplicate Document Detection.
b) Illustrate variable length index computer. [5+5]

10. Write a detailed note on Semi-Structured Search using a Relational Schema. [10]

OR

11.a) Demonstrate user-defined operations in SQL.
b) Discuss the relations which model the document collections. [5+5]

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